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Your Roll No. .....

CEN-503

B.Tech.(Computer Engineering) V Semester Examination 2023

Microprocessor

Paper No. : CEN-503

Time: 3 Hours

Maximum Marks: 45

(Write your Roll No. on top immediately on receipt of this question Paper)

Note: - Attempt all questions.

CO1

Q1. (a). Draw and explain functional block diagram of 8086 microprocessor. 4.5

(b). Discuss the various pins of 8085 Microprocessor after drawing its pin diagram. 4.5

OR

Q1'(a) Draw and explain timing diagram of OUT 26 H instruction of 8085 stored at address 2400H. 4.5

Q1'(b)Distinguish between 8085 Microprocessor and 8086 Microprocessor 4.5

CO2

2(a). Explain the various flags available in 8085 microprocessor with their significance and their setting/ resetting with suitable example.

2(b) Explain the meanings of the following instructions of 8085/8086 after writing their syntax. Also illustrate their use with the help of suitable example. 4.5

(i)LHLD

(ii)DAD

(iii)LOOP

(iv)MOVSB

CO3

3(a). Write a well documented program in assembly language of 8085 to convert 8 bit binary number to decimal number. 4.5

3(b). Write a well documented program in the assembly language of 8085 to find factorial of a given integer 4.5

OR

3'(a).Write a program in assembly language of 8085 to arrange 350 integers stored at consecutive memory locations starting with address 2242H in Descending order. Program should be well documented. 4.5

3'(b Write a program in assembly language of 8085 to generate Modulo-09 counter with a delay of 60 ms between each count assuming a system with 0.5 microsecond clock period. Show all calculations. 4.5

CO4

4(a). Explain how I/O devices are interfaced as I/O Mapped I/O with help of suitable diagram and examples. 4.5

4(b). Explain how RIM and SIM instructions are used for interrupts in 8085 Microprocessor with the help of suitable example. 4.5

CO5

5. (a). Write a program in assembly language of 8086 to find out the number positive and no of negative numbers out of N 16 bit numbers stored at consecutive memory locations. 4.5

5(b). Discuss the various addressing modes of 8086 Microprocessor with the help of suitable example. 4.5